Wednesday Lab Examp Set A

Q1 Given an array A containing distinct integers and an integer x present in the array Arr. Rearrange the elements of the array such that all the elements that are smaller than x are placed before x, followed by all the elements that are larger than x. Note that the final array does not need to be sorted. You cannot define any additional arrays (other than the given array) for this question.

Example :Input: Enter the number of elements in the array: 7
Enter element of array : 77 4 34 12 9 45 1
Enter value of x : 12
Output: 1 9 4 12 77 45 34 (Please note that the solution may not be unique. There may be various correct.)

Q2. You are given an unsorted array of integers. Write a program in C to remove the duplicate elements in-place such that each element appears only once. The program should also return the new length of the modified array. You must modify the array in-place without using any extra space (like an auxiliary array or set or sorting is not allowed). And the sequence of the input should be the same. It should not change after removing the duplicate value.

Example :Input: nums = {3, 1, 2, 3, 4, 1, 2, 5}
n = 8
Output: New length: 5
Modified array: 3 1 2 4 5

Q3. Given arrival and departure times of all trains that reach a railway station. Find the minimum number of platforms required for the railway station so that no train is kept waiting. Consider that all the trains arrive on the same day and leave on the same day. Arrival and departure time can never be the same for a train but we can have arrival time of one train equal to departure time of the other. At any given instance of time, same platform can not be used for both departure of a train and arrival of another train. In such cases, we need different platforms.

Example 1: Input: n = 6
arr[] = 0900, 0940, 0950, 1100, 1500, 1800
dep[] = 0910, 1200, 1120, 1130, 1900, 2000
Output: 3 Explanation: Minimum 3 platforms are required to safely arrive and depart all trains.
Example 2: Input: n = 3
arr[] = 0900, 1100, 1235
dep[] = 1000, 1200, 1240
Output: 1 Explanation: Only 1 platform is required to safely manage the arrival and departure of all trains.
Your Task: You don't need to read input or print anything. Your task is to complete the function findPlatform() which takes the array arr[] (denoting the arrival times), array dep[] (denoting the departure times) and the size of the array as inputs and returns the minimum number of platforms required at the railway station such that no train waits. Note : Time intervals are in the 24-hour format(HHMM) , where the first two characters represent hour (between 00 to 23) and the last two characters represent minutes

Wednesday Lab Examp Set B

Q1 Two friends like to pool their money and go to the ice cream parlour. They consistently choose two distinct flavors, and they try to spend all of their money. Given an array A containing prices for different flavors of ice cream, select the two that will cost all the money they have. Print the indices of the flavors of ice-cream that they will choose. The following constraints are applicable It is guaranteed that they will be able to spend all their money Nested loops are not allowed.

Example 1: Input: Total money they have:4
Total flavour of ICE cream: 5
Price of ICE creams :1 4 5 3 2
Output: index of ICE cream bought 0 3
Explanation: A[0] + A[3] = 4

Example 2: Input: Total money they have 4
Total flavour of ICE cream: 4
Price of ICE creams:2 2 4 3
Output: index of ICE cream bought :0 1
Explanation: A[0] + A[1] = 4

Q2 Given an integer n, write a function that returns an array ans of length n+1n + 1n+1 such that for each iii (where 0≤i≤n), ans[i] contains the number of 1's in the binary representation of i.
Example 1: Input: n = 2
Output: [0,1,1]
Explanation:0 --> 0 | 1 --> 1 | 2 --> 10
Example 2: Input: n = 5
Output: [0,1,1,2,1,2] Explanation: 0 --> 0| 1 --> 1 | 2 --> 10 | 3 --> 11| 4 --> 100 | 5 --> 1010

Q3 You are a robber planning to rob houses along a street. Each house has a certain amount of money stored in it. However, you cannot rob two adjacent houses because they have security systems that will alert the police if you do.

Your task: Given an array of integers nums, where each integer represents the amount of money in each house, determine the maximum amount of money you can rob without alerting the police.
Examples:
Input: nums = [1, 2, 3, 1]
Output: 4
Explanation: You can rob house 1 (money = 1) and house 3 (money = 3). The total amount you can rob is 1 + 3 = 4.

Input: nums = [2, 7, 9, 3, 1]
Output: 12
Explanation: You can rob house 1 (money = 2), house 3 (money = 9), and house 5 (money = 1). The total amount you can rob is 2 + 9 + 1 = 12.